00029 **INSST/TDD/V/25/7**

STATISTICS

(Paper—III)

Full Marks : 100

Time : 3 hours

The figures in the margin indicate full marks for the questions

This paper is in two Parts. Attempt both Parts

PART—1

Answer *any* **five** questions

1. *(a)* Let $X \sim \text{Bin}(2, p)$ and $Y \sim \text{Bin}(4, p)$, where X and Y are independent random variables and $0 < p < 1$. If

$$P(X \geq 1) = \frac{5}{9}$$

then find $P(Y \geq 1)$. 4

(b) A man is dealt 4 Spade cards from an ordinary pack of 52 cards. If he is given one more card at random, find the probability that this additional card is not a Spade. 3

(c) The probability that a student passes a test in physics is $\frac{2}{3}$. Moreover, the probability that he passes both the tests

in Physics and in English is $\frac{14}{45}$. Additionally, it is also given that the probability that he passes at least one of the tests is $\frac{4}{5}$. What is the probability that he passes the test in English? 3

2. *(a)* Obtain the mean and variance of Poisson distribution. 4

(b) The average salary of male employees in a firm was ₹ 5,200 and that of female employees was ₹ 4,200. The average salary of all the employees was ₹ 5,000. Find the percentage of male and female employees. 3

(c) There are three candidates for the post of principal of a girls' college viz. Mr. Chatterji, Mr. Ayengar and Mr. Singh. Their chances of being appointed are in the proportion 4 : 2 : 3, respectively.

If Mr. Chatterji is selected, then the probability that he would introduce co-education in the college is 0·3.

Similarly, the probabilities of Mr. Ayengar and Mr. Singh introducing co-education are 0·5 and 0·8, respectively.

What is the probability that co-education will be introduced in the college? 3

3. (a) Let X and Y be i.i.d. $N(1, 1)$ variates. Find $P(2X + 4Y \le 6)$. 4

(b) The diameter, say X, of an electric cable is assumed to be a continuous random variable with p.d.f.

$$f(x) = \begin{cases} kx(1-x), & 0 < x < 1 \\ 0, & \text{otherwise} \end{cases}$$

(i) Find k.

(ii) Obtain the distribution function of X.

(iii) Find $P\left(X \le \frac{1}{2}\right)$. 2+2+2=6

4. (a) Find the maximum likelihood estimate (MLE) of the parameter λ of a Poisson distribution on the basis of a sample of size n. Also find its variance. 4

(b) Let X_1, X_2 and X_3 be a random sample of size 3 from a population with mean μ and variance σ^2. Let T_1, T_2 and T_3 be estimators, where

$$T_1 = X_1 + X_2 - X_3$$
$$T_2 = 2X_1 + 3X_3 - 4X_2$$
$$T_3 = (kX_1 + X_2 + X_3)/3$$

(i) Find the value of k so that T_3 is an unbiased estimator of μ.

(ii) Find $\text{var}(T_1)$, $\text{var}(T_2)$ and $\text{var}(T_3)$, and indicate which is the best estimator of μ in the sense of minimum variance. 2+4=6

5. Define and briefly explain the following : 2×5=10

(a) Type 1 error

(b) Type 2 error

(c) Critical region

(d) Null hypothesis

(e) Composite hypothesis

6. (a) Obtain the characteristic equation of the following matrix :

$$A = \begin{bmatrix} 2 & -1 & 1 \\ -1 & 2 & -1 \\ 1 & -1 & 2 \end{bmatrix}$$

Also obtain A^{-1}. 4

(b) Find the rank of the following matrix by reducing it to canonical form : 3

$$B = \begin{bmatrix} 1 & 2 & 1 & 0 \\ -2 & 4 & 3 & 0 \\ 1 & 0 & 2 & -8 \end{bmatrix}$$

(c) (i) Define correlation coefficient.

(ii) State the two lines of regression. 1+2=3

PART—2

Answer *any* **five** questions

7. (a) State the advantages and disadvantages of randomised block design (RBD). 4

(b) What are the three principles of design of experiments? Explain them briefly. 3

(c) Explain how the three principles of design are implemented in RBD. 3

8. (a) Show that in SRSWOR, the variance of the sample mean is given by

$$\text{var}(\bar{y}_n) = \frac{N-n}{Nn} S^2$$ 4

(b) With usual notations, show that

$$\text{var}(\bar{y}_{\text{st}})_{\text{Ney}} \le \text{var}(\bar{y}_{\text{st}})_p$$ 6

9. (a) (i) Define Laspeyre's and Paasche's price index number.

(ii) If $L(p)$ and $P(q)$ represent respectively the Laspeyre's index number for prices and Paasche's index number for quantities, show that

$$\frac{L(p)}{L(q)} = \frac{P(p)}{P(q)}$$ 2+2=4

(b) Show that Fisher's index number satisfies the time-reversal test. 3

(c) What is cost of living index number? Briefly describe a method for its construction. 1+2=3

10. (a) With the data given below, complete the life table of the population of a certain type of insects, x being the age in days and $l_x = 1000$ for $x = 0$: 4

x :	0	1	2	3	4	5	6	7	8
q_x :	0·120	0·005	0·010	0·050	0·100	0·500	0·800	0·900	0·950

(b) Compute—

(i) GFR (General Fertility Rate)

(ii) TFR (Total Fertility Rate)

(iii) GRR (Gross Reproduction Rate)

from the data given below (assume that the proportion of female births is 46·2 percent) :

Age group	*Number of women* (in '000)	*Total birth*
15–19	16·0	260
20–24	16·4	2244
25–29	15·8	1894
30–34	15·2	1320
35–39	14·8	916
40–44	15·0	280
45–49	14·5	145

2+2+2=6

11. *(a)* Distinguish between parametric and non-parametric tests. Write down the advantages and disadvantages of non-parametric tests. 4

(b) Consider a Markov chain with transition probability matrix

$$P = \begin{pmatrix} 0 & 0 & 0\cdot5 & 0 & 0\cdot5 \\ 0 & 0\cdot5 & 0 & 0\cdot5 & 0 \\ 0\cdot5 & 0 & 0 & 0 & 0\cdot5 \\ 0 & 0\cdot5 & 0 & 0\cdot5 & 0 \\ 0\cdot5 & 0 & 0\cdot5 & 0 & 0 \end{pmatrix}$$

Show that the chain is aperiodic. 3

(c) Describe the different components of a time series. 3

12. *(a)* Why do we use control chart for attributes? In this context, define p-chart and c-chart. 4

(b) If the third differences are constant, prove that

$$\int_{-1}^{1} f(x)\,dx = \frac{2}{3}\left[f(0) + f\left(\frac{1}{\sqrt{2}}\right) + f\left(-\frac{1}{\sqrt{2}}\right)\right] \qquad 3$$

(c) Define Hotelling T^2-statistic. Discuss its use. 1+2=3

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